

Lab: Python Preliminaries

ACTL3143 & ACTL5111 Deep Learning for Actuaries

Getting setup

You can use Google Colaboratory to run Python. Eventually you'll want to run Python on your own computer though. In Week 1, follow along this video to get everything installed to run Python on your home computer.

<https://youtu.be/tSRKB1sLsGE>

Python Basics

In this module we will be introducing you to Python, covering fundamentals such as variables, control flow, functions, classes, and packages. You should all be familiar with the language R. R shares many similarities with Python, including some syntax, data types, and uses - R and Python are the two most popular languages for data science [source: edx.org]. Because of this, you should be able to easily understand the basics of Python, and in this lab, we will make some references to R.

Variables and basic types

Much like with R, Python can also be used as a calculator:

```
# Addition  
1 + 1
```

2

```
# Subtraction  
9 - 6
```

3

```
# Multiplication  
12 * 8
```

96

```
# Division  
54 / 18
```

3.0

```
# Modulo  
39 % 8
```

7

```
# Brackets  
3 * (4 + 5)
```

27

```
# Powers - note that unlike R, Python uses "**" to denote powers  
4 ** 3
```

64

Assigning values to variables is easy:

```
a = 1  
print(a)
```

1

```
a = 2  
b = 3  
b = a  
print(b)
```

2

Types of variables in Python include:

- int
- float
- string
- bool
- NoneType

You can check the type of a variable by using the `type()` function:

```
print(type(1))
print(type(1.0))
print(type("Hello World!"))
print(type(1 == 1.0))
print(type(None))
```

```
<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
<class 'NoneType'>
```

Shorthand assignments

```
x = 1
x += 2
print(x)
```

3

```
x = 1
x -= 2
print(x)
```

-1

```
x = 1
x *= 2
print(x)
```

2

```
x = 1
x /= 2
print(x)
```

0.5

Strings

Much like in R, you can encode strings by using single or double quotation marks:

```
s1 = "Hello"
```

```
s2 = 'World!'
```

You can concatenate strings by using the `+` operator:

```
s1 + " " + s2
```

```
'Hello World!'
```

F-strings are one of the ways you can substitute the values of variables into a string:

```
f"{s1} {s2}"
```

```
'Hello World!'
```

Logical operators

In Python, the logical operators **and**, **or**, and **not** are used. This is unlike in R, in which the symbols `&`, `|`, and `!` are used.

```
True and False
```

False

```
True or False
```

True

Converting types

Use the functions `int()`, `float()`, and `str()` to convert values from one type to another:

```
x = 3
x = str(x)
print(x)
```

3

Exercises

1. Calculate the value of $5 * (2 + 4^3)$ using Python.
2. In R, you can use the function `is.string()` to determine whether a value is a string. Unfortunately, there isn't a similar function in Python. Can you figure out a way to check whether a value is a string in Python? What about an integer, or a bool?
3. What is the value of `type(type(bool(int("3"))))`?
4. What is the value of `not(True and not False or False and not (True or False))`?

```
5*(2+4**3)
```

330

```
x = "Hello"
print(x == str(x))
```

True

```
type(type(bool(int("3"))))
```

type

```
not(True and not False or False and not (True or False))
```

False

Data Structures

There are four types of built-in data structures available in Python: lists, dictionaries, tuples, and sets.

Lists are very much like R vectors, in which they hold a series of values of any type. Lists in Python can be altered, whether it be by deleting or adding elements, or by editing specific elements inside the list. This feature is known as **mutability**. You will later see that some of the other Python data structures do not have this feature.

```
l = [1,2,3,4,5]
```

Tuples are similar to lists, except that they are **immutable**, i.e. they cannot be altered.

```
t = (1,3)
```

Dictionaries are a type of data structure that stores data in key-value pairs.

Dictionaries are mutable, however, you cannot change the name of a key.

```
wam_dict = {  
    "Alice": 87.4,  
    "Bob": 77.9,  
    "Charlie": 81.3  
}
```

Sets are a type of **unordered** data structure that store a collection of **unique** values. You cannot change the specific values inside a set, but you can remove and insert elements.

```

fruits = {"Apple", "Banana", "Strawberry"}
print(fruits)

fruits.add("Mango")
print(fruits)

fruits.add("Banana")
print(fruits) #This will not return an error, but the duplicate element will not be added

fruits.remove("Apple")
print(fruits)

```

```

{'Apple', 'Strawberry', 'Banana'}
{'Apple', 'Mango', 'Strawberry', 'Banana'}
{'Apple', 'Mango', 'Strawberry', 'Banana'}
{'Mango', 'Strawberry', 'Banana'}

```

Exercises

- a) This exercise will get you familiar with Python's built-in list methods.
 1. Create an empty list and assign it to the variable `l`.
 2. Append the numbers 7, 2, and 11 to the list.
 3. Sort the list using the `sort()` method.
 4. Insert the number 8 at index 1 using the `insert()` method.
 5. Reverse the list using `reverse()`.
 6. Call `pop()` on the list twice.
 7. Use `remove()` to remove the number 11 from the list.
 8. Print the list.
 9. Clear the list.
- b) This exercise will teach you how to build a dictionary by combining a list and a tuple. This can also be done with two lists or two tuples.
 1. Create a tuple containing the values "Circle", "Triangle", and "Square". Call this tuple `shapes`.
 2. Create a list containing the values 1, 3, and 4. Call this list `sides`.
 3. Combine `shapes` and `sides` using `dict(zip())`. Print the dictionary.
 4. Add "Pentagon": 5 to this dictionary to show that it is mutable. Print the dictionary.

If-else statements

In Python, note the use of `elif` rather than `else if`. Also, note that Python uses indentation (as opposed to braces in R, C++, JavaScript, and other languages) to denote different blocks of code.

```
profit = -30

if profit >= 0:
    print("Profitable")
elif profit == 0:
    print("Break even")
else:
    print("Loss")
```

Loss

For loops

You can use for loops in Python to iterate through each value in a list, tuple, or other collection.

```
actl_core = ["ACTL1101", "ACTL2111", "ACTL2131", "ACTL2102"]

for course in actl_core:
    print(course)
```

ACTL1101
ACTL2111
ACTL2131
ACTL2102

You can also use for loops to iterate through a sequence of numbers by creating the `range()` object.

```
for i in range(5):
    print(i ** 2)
```


0
1
4
9
16

While loops

while loops can be used to perform the same tasks as for loops, however, they tend to be more cumbersome to put together.

```
i = 0

while i < 5:
    print(i ** 2)
    i += 1
```

0
1
4
9
16

while loops are useful, however, for creating “sentinel” loops, in which a loop runs until a variable reaches a specified value called a sentinel:

```
scores = [9, 9, 6, 7, 3, 10, 0, 1, 4, 6, 2]
print(sum(scores))

i = 0
pre_fail_score = 0

while scores[i] != 0: # 0 is a sentinel value
    pre_fail_score += scores[i]
    i += 1

print(pre_fail_score)
```

57
44

Exercises

1. Recreate the FizzBuzz program - a classic task in coding interviews - in Python. FizzBuzz is a children's game in which people in a circle count to 100 - with a twist. Your FizzBuzz program should loop from 1 to 50, and print, on a new line:
 - "Fizz!" if the current number is divisible by 3
 - "Buzz!" if the current number is divisible by 5
 - "FizzBuzz!" if the current number is divisible by both 3 and 5
 - Otherwise, print the current number.
2. This exercise will introduce you to the **break** and **continue** statements in Python.
 - Create a new list with the numbers 3, 7, -9, 11, 2, -5, 0, 4.
 - Create a variable, **sum_pos**, and set it equal to 0.
 - Create a for loop that iterates through each of the numbers in the list.
 - If the current number is positive, add it to **sum_pos**.
 - If the current number is negative, ignore it and move to the next number using the **continue** statement.
 - If the current number is zero, stop the loop using **break**.

Functions

Functions are instantiated using the **def** keyword.

```
def addOne(x = 0):  
    return x + 1  
  
print(addOne(10))  
print(addOne()) #Using default arguments
```

11
1

Exercises

1. Build a function, **factorial(n)**, that takes in one input **n**, and returns **n!**.
2. Build a function, **fibonacci(n)**, that takes in one input **n**, and returns the **nth** Fibonacci number.

Data Science Libraries

A couple of fundamental data science packages in Python are NumPy and Pandas. NumPy is a package for handling matrices and vector math, while Pandas handles dataframes and data wrangling.

Libraries are imported using the `import` keyword:

```
import numpy
```

You can set an alias to the libraries you are importing. Usually this is done to simplify the name of a long library.

```
import numpy as np
import pandas as pd
```

You can also import specific functions from a library by using the `from` keyword:

```
from sklearn.preprocessing import StandardScaler
```

In this lab, we will be working with two libraries used for data processing, NumPy and Pandas.

A Note on Installing Libraries

If you have successfully installed Anaconda onto your system, you should already have NumPy and Pandas installed as well. However, if for some reason you do not have a particular library installed, or you would like to update a particular library, you can use the command line to install new packages.

You can either open up Command Prompt/Terminal and type:

```
pip install numpy
```

The `pip` method will also work on Anaconda Prompt. This will install the libraries onto your machine. When installing libraries, it is highly recommended that you create a Conda **environment**, as this allows you to install and manage separate sets of libraries for each Python project you are working on.

For a tutorial on how to set up your own environments, see <https://docs.conda.io/projects/conda/en/latest/user-guide/concepts/environments.html>

NumPy

NumPy is a package used for scientific computing in Python, with the ability to perform advanced mathematical operations, linear algebra, and vectorisation. Core to the NumPy package is the NumPy array.

NumPy 1D arrays

Unlike lists in base Python, NumPy arrays can only work with numerical data. NumPy arrays are also faster and consumes less memory than Python lists (source: numpy.org/doc/stable/user/absolute_beginners.html).

```
l1 = [1,1,1]
l2 = [2,2,2]

a1 = np.array(l1)
a2 = np.array(l2)

#What do you notice?
print(l1 + l2)
print(a1 + a2)
```

```
[1, 1, 1, 2, 2, 2]
[3 3 3]
```

As you can see in the above code snippet, NumPy arrays are designed for linear algebra operations.

Other operations you can do include adding and multiplying arrays by a constant, calculating determinants of matrices, and even calculating eigenvalues and eigenvectors:

```
a1 + 3 #adds 3 to each element of the array, returns an error if done to a list
a1 * 3 #multiplies each element by 3
```

```
array([3, 3, 3])
```

```
m1 = np.array([[2,4],[1,3]]) #creating a 2D array, i.e. a matrix
print(m1)

print(np.linalg.det(m1)) #Determinant
print(np.linalg.eig(m1)) #Eigenvalues and eigenvectors
```

```

[[2 4]
 [1 3]]
2.0
EigResult(eigenvalues=array([0.43844719, 4.56155281]), eigenvectors=array([[ -0.93153209, -0.93153209]
 [ 0.36365914, -0.5392856 ]]))

```

You can create arrays using ranges or linearly spaced sequences:

```

array_range = np.arange(5)
array_lin = np.linspace(start = 0, stop = 1, num = 6)

print(array_range)
print(array_lin)

```

```

[0 1 2 3 4]
[0.  0.2 0.4 0.6 0.8 1. ]

```

NumPy 2D arrays

As mentioned beforehand, you can create a matrix by feeding a list of lists into `np.array()`:

```

m1 = np.array([[2,4],[1,3]])

```

You can also create matrices of zeroes and identity matrices:

```

m_zero = np.zeros([3,3]) #3 x 3 matrix
print(m_zero)

m_ones = np.ones([3,3])
print(m_ones)

m_id = np.identity(3)
print(m_id)

```

```

[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]

```

```
[[1. 0. 0.]  
 [0. 1. 0.]  
 [0. 0. 1.]]
```

Pandas

Pandas is a Python library used for working with tabular data. It contains tools for data manipulation, time series, and data visualisation. Pandas can be considered a Python equivalent to `dplyr`, and core to Pandas is the `DataFrame` object, which is analogous to R's `data.frame` type.

```
import pandas as pd
```

DataFrames

For this lab we will be working with the Titanic machine learning dataset - a legendary dataset in the data science community. It is available at <https://www.kaggle.com/competitions/titanic/data>, and we will specifically be using `train.csv`.

To use the dataset in Google Colab, we need to upload and then import it. To see which datasets are available in Google Colab, click the folder icon on the sidebar. Here, you can see the datasets you have uploaded, as well as any sample datasets that are already built into Google Colab. To upload files, click the upload icon that appears and select the file that you want to upload.

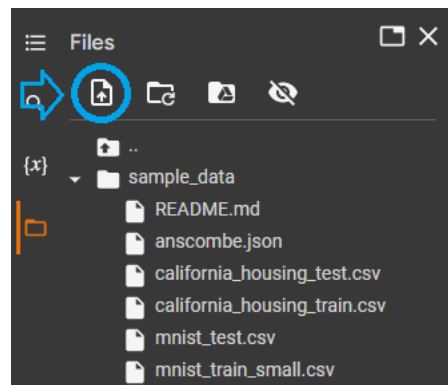


Figure 1: Google Colab Files

We will import the dataset using Pandas' `read_csv()` function.

```
titanic = pd.read_csv("train.csv")
```

This creates a `DataFrame` object, which is a 2-dimensional, tabular data structure.

There are a number of methods available in Pandas to inspect your data, including `.head()` and `.info()`.

```
titanic.head() #much like the head() function in R, this method prints the first 5 rows of t
```

	PassengerId	Survived	Pclass	Name	Sex	Age	Sib
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0

```
titanic.tail(10) # Prints last 10 rows
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Pa
881	882	0	3	Markun, Mr. Johann	male	33.0	0	0
882	883	0	3	Dahlberg, Miss. Gerda Ulrika	female	22.0	0	0
883	884	0	2	Banfield, Mr. Frederick James	male	28.0	0	0
884	885	0	3	Sutehall, Mr. Henry Jr	male	25.0	0	0
885	886	0	3	Rice, Mrs. William (Margaret Norton)	female	39.0	0	5
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0

```
titanic.info() # Gives a list of columns, their counts and their types, akin to the str() fu
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null   int64
```

```

1   Survived      891 non-null   int64
2   Pclass       891 non-null   int64
3   Name         891 non-null   str
4   Sex          891 non-null   str
5   Age         714 non-null   float64
6   SibSp        891 non-null   int64
7   Parch        891 non-null   int64
8   Ticket       891 non-null   str
9   Fare         891 non-null   float64
10  Cabin        204 non-null   str
11  Embarked     889 non-null   str
dtypes: float64(2), int64(5), str(5)
memory usage: 118.9 KB

```

Selecting columns of a Pandas DataFrame is done using square brackets notation:

```
titanic["Age"] # Selecting "Age" column from dataset
```

```

0      22.0
1      38.0
2      26.0
3      35.0
4      35.0
...
886     27.0
887     19.0
888      NaN
889     26.0
890     32.0
Name: Age, Length: 891, dtype: float64

```

```
titanic[["Sex","Age"]] # Selecting multiple columns
```

	Sex	Age
0	male	22.0
1	female	38.0
2	female	26.0
3	female	35.0
4	male	35.0
...	...	...

	Sex	Age
886	male	27.0
887	female	19.0
888	female	NaN
889	male	26.0
890	male	32.0

There are several ways of selecting rows in a DataFrame, including selecting by row number using the square bracket notation or the `.iloc` method, or selecting by row name using the `.loc` method.

```
titanic[4:9] # Selecting rows by the index (can be different to row number)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	Survived
4	5	0	3	Allen, Mr. William Henry	male	35.0	0
5	6	0	3	Moran, Mr. James	male	NaN	0
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0

```
titanic.iloc[4:9] # Selecting rows by their row numbers
```

	PassengerId	Survived	Pclass	Name	Sex	Age	Survived
4	5	0	3	Allen, Mr. William Henry	male	35.0	0
5	6	0	3	Moran, Mr. James	male	NaN	0
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0

```
titanic.set_index("Name", inplace=True) #sets the "Name" column as the index
# By setting inplace = True, we modify the existing DataFrame rather than creating a new one
# In other words, we do not need to assign it back to the titanic variable.
```

```
# Selecting rows using .loc
titanic.loc[["Allen, Mr. William Henry", "Moran, Mr. James"]]
```

	PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare
Name									
Allen, Mr. William Henry	5	0	3	male	35.0	0	0	373450	8.0500
Moran, Mr. James	6	0	3	male	NaN	0	0	330877	8.4583

When selecting both rows and columns, using `.loc` or `.iloc` is necessary:

```
titanic.iloc[4:9, [0, 3]] # Selecting rows 4 to 8, and columns 0 and 3
```

	PassengerId	Sex
Name		
Allen, Mr. William Henry	5	male
Moran, Mr. James	6	male
McCarthy, Mr. Timothy J	7	male
Palsson, Master. Gosta Leonard	8	male
Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	9	female

```
titanic.loc[["McCarthy, Mr. Timothy J", "Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)"]]
```

```
Name
McCarthy, Mr. Timothy J    54.0
Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)    27.0
Name: Age, dtype: float64
```

You can use the bracket notation to filter the dataset:

```
titanic[titanic["Age"] >= 18]
```

	PassengerId	Survived	Pclass	Sex	Age	SibSp
Name						
Braund, Mr. Owen Harris	1	0	3	male	22.0	1
Cumings, Mrs. John Bradley (Florence Briggs Thayer)	2	1	1	female	38.0	1
Heikkinen, Miss. Laina	3	1	3	female	26.0	0
Futrelle, Mrs. Jacques Heath (Lily May Peel)	4	1	1	female	35.0	1
Allen, Mr. William Henry	5	0	3	male	35.0	0
...	...	...	...	...	...	...

Name	PassengerId	Survived	Pclass	Sex	Age	SibSp
Rice, Mrs. William (Margaret Norton)	886	0	3	female	39.0	0
Montvila, Rev. Juozas	887	0	2	male	27.0	0
Graham, Miss. Margaret Edith	888	1	1	female	19.0	0
Behr, Mr. Karl Howell	890	1	1	male	26.0	0
Dooley, Mr. Patrick	891	0	3	male	32.0	0

This has reduced the dataset from 891 rows to 601.

If we wanted to combine multiple conditions together, we can use conditional operators. However, Python's usual conditional operators (`and`, `or`, `not`) will not work here, and instead we will need to use symbols (`&`, `|`, `!`).

```
# Selecting passengers whose ages are 18 and above and are in passenger class 3.
titanic[(titanic["Age"] >= 18) & (titanic["Pclass"] == 3)] #Note that we need to wrap each c
```

Name	PassengerId	Survived	Pclass	Sex	Age	SibSp
Braund, Mr. Owen Harris	1	0	3	male	22.0	1
Heikkinen, Miss. Laina	3	1	3	female	26.0	0
Allen, Mr. William Henry	5	0	3	male	35.0	0
Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	9	1	3	female	27.0	0
Saunderscock, Mr. William Henry	13	0	3	male	20.0	0
...	...	...	...	...	...	...
Markun, Mr. Johann	882	0	3	male	33.0	0
Dahlberg, Miss. Gerda Ulrika	883	0	3	female	22.0	0
Sutehall, Mr. Henry Jr	885	0	3	male	25.0	0
Rice, Mrs. William (Margaret Norton)	886	0	3	female	39.0	0
Dooley, Mr. Patrick	891	0	3	male	32.0	0

That line of code is quite longwinded, so if you wanted to filter your DataFrame in a more concise way, you can use the `.query()` method:

```
titanic.query("Age >= 18 & Pclass == 3")
```

Name	PassengerId	Survived	Pclass	Sex	Age	SibSp
Braund, Mr. Owen Harris	1	0	3	male	22.0	1
Heikkinen, Miss. Laina	3	1	3	female	26.0	0
Allen, Mr. William Henry	5	0	3	male	35.0	0
Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	9	1	3	female	27.0	0
Saunderscock, Mr. William Henry	13	0	3	male	20.0	0
...	...	...	...	...	...	...
Markun, Mr. Johann	882	0	3	male	33.0	0
Dahlberg, Miss. Gerda Ulrika	883	0	3	female	22.0	0
Sutehall, Mr. Henry Jr	885	0	3	male	25.0	0
Rice, Mrs. William (Margaret Norton)	886	0	3	female	39.0	0
Dooley, Mr. Patrick	891	0	3	male	32.0	0

In Pandas you can aggregate datasets using the `.groupby()` method:

```
titanic.groupby("Pclass").sum()["Survived"]
```

Pclass

1 136

2 87

3 119

Name: Survived, dtype: int64

Notice in the above line of code, we combined two methods. In Pandas, you can chain multiple methods together, much like dplyr's pipeline operator (`%>%`) in R.

```
# Select the names of passengers in class 3 who are 65 years of age or older.
titanic.reset_index().query("Pclass == 3 & Age >= 65")["Name"]
```

116 Connors, Mr. Patrick

280 Duane, Mr. Frank

851 Svensson, Mr. Johan

Name: Name, dtype: str

Exercises

1. Filter the dataset to people where **Embarked** is Q.
2. Filter the dataset to people 18 years or older, and **Fare** is less than 10
3. Filter the dataset to people with an above-median age.
4. What is the highest value of **Fare** for female passengers in class 2?

Series

Let's select the Ticket column:

```
titanic["Ticket"]
```

```
Name
Braund, Mr. Owen Harris          A/5 21171
Cumings, Mrs. John Bradley (Florence Briggs Thayer)  PC 17599
Heikkinen, Miss. Laina          STON/O2. 3101282
Futrelle, Mrs. Jacques Heath (Lily May Peel)        113803
Allen, Mr. William Henry        373450
...
Montvila, Rev. Juozas           211536
Graham, Miss. Margaret Edith    112053
Johnston, Miss. Catherine Helen "Carrie"            W./C. 6607
Behr, Mr. Karl Howell           111369
Dooley, Mr. Patrick             370376
Name: Ticket, Length: 891, dtype: str
```

When selecting a single column of the DataFrame, Pandas returns what is known as a **Series**. This is a data structure used to represent one-dimensional data, much like a list or NumPy array. They are more flexible than NumPy arrays because they can hold non-numeric data types. However, they are not as flexible as lists because they can only hold one datatype at a time. If you try to create a Series with values of different data types, Pandas will convert all the elements of the Series into strings.

```
# You can create series from lists, tuples, and NumPy arrays
l = ["The", "quick", "brown", "fox"]
t = (3,1,4,1,5,9)
a = np.array(t)
mix = ["this", 3, "will", True, "convert"]

print(pd.Series(l))
print(pd.Series(t))
print(pd.Series(a))
print(pd.Series(mix)) #converted into strings
```

```
0    The
1  quick
2  brown
```

```

3      fox
dtype: str
0      3
1      1
2      4
3      1
4      5
5      9
dtype: int64
0      3
1      1
2      4
3      1
4      5
5      9
dtype: int64
0      this
1          3
2      will
3      True
4      convert
dtype: object

```

Matplotlib is a Python library for creating high-quality data visualisations. It can be used to build a wide variety of charts, and in this tutorial we will explore how to build line plots, scatter plots, bar plots, and histograms. Charts built using Matplotlib are highly customisable.

As a data scientist, the ability to visualise your data effectively is important as it allows you to develop a deep understanding and relationship with your data. You'll be able to see potential trends and data characteristics that you can incorporate or account for in your modelling later.

Installing Matplotlib

You should already have Matplotlib installed on your computer if you are using Python through Anaconda. It is also installed by default if you are using Google Colab. However if for some reason you don't have the library installed yet, you can do so using `pip`. Open up Command Prompt/Terminal and type in:

```
pip install matplotlib
```

You can also use the `!pip` keyword to install it directly into your notebook, or install it using Anaconda Navigator.

Once Matplotlib is installed, you can import it into your Python program:

```
import matplotlib.pyplot as plt
```

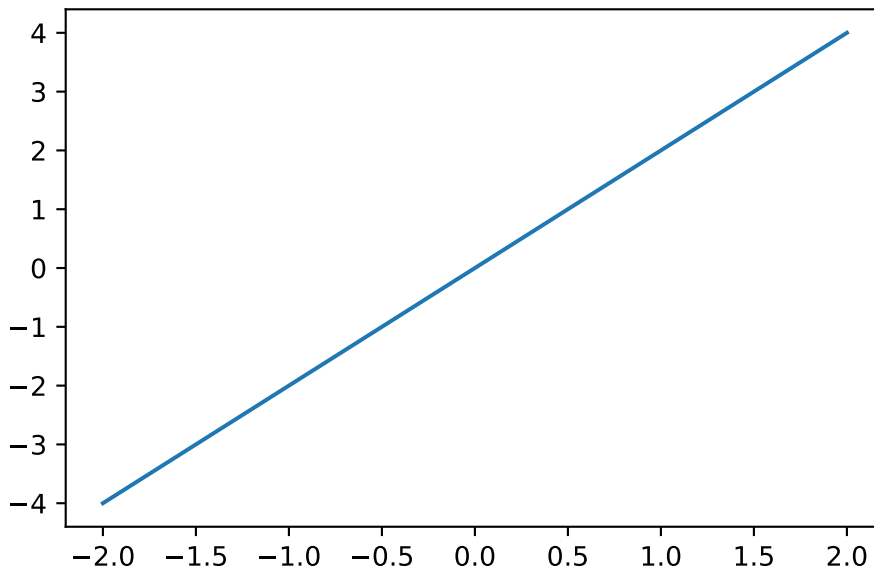
Note that we specifically need to import `pyplot` as opposed to Matplotlib itself. This is because `pyplot` is an interface for Matplotlib that enables the library to work more like MATLAB, in which you will first initialise the figure and then each function makes some change to that figure (source: <https://matplotlib.org/stable/tutorials/introductory/pyplot.html>).

Basic plot types

Line plot

Pyplot's `plot()` function will create a line plot:

```
# Create sample data  
x = [-2,-1,0,1,2]  
y = [-4,-2,0,2,4]  
  
# Create line plot  
plt.plot(x,y)
```



As you can see, we have created a simple line plot. We can customise this by adding a title, customising the x- and y-axes, and even changing the colour of the line:

```

# Create sample data
x = [-2,-1,0,1,2]
y = [-4,-2,0,2,4]

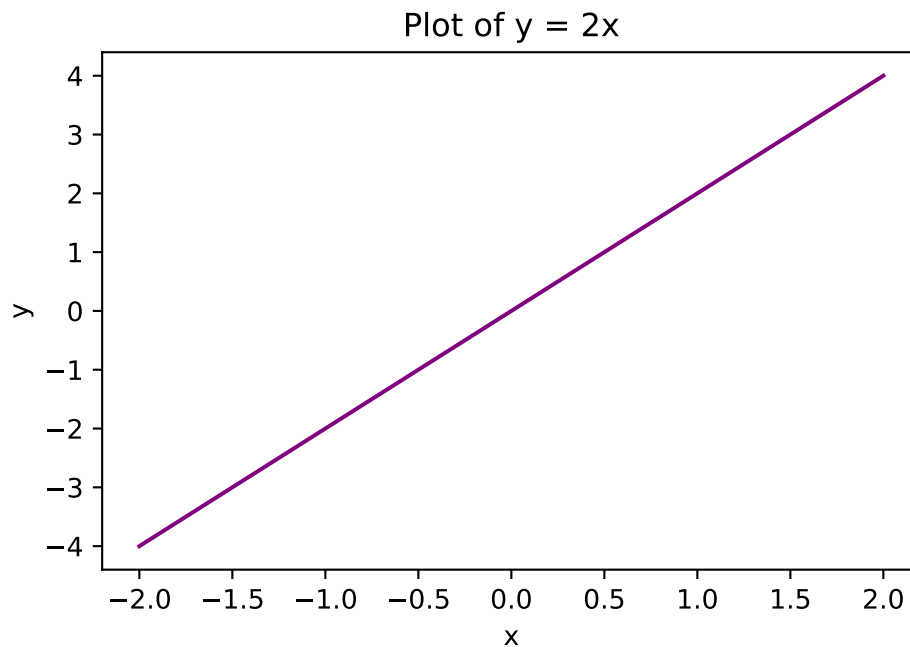
# Create line plot
plt.plot(x,y, color = "purple")

# Add title
plt.title("Plot of  $y = 2x$ ")

# Add axes labels
plt.xlabel("x")
plt.ylabel("y")

```

Text(0, 0.5, 'y')



You can also add multiple lines to a plot:

```

# Create sample data
x = [-2,-1,0,1,2]
y1 = [-4,-2,0,2,4]
y2 = [6,3,0,-3,-6]

```



```

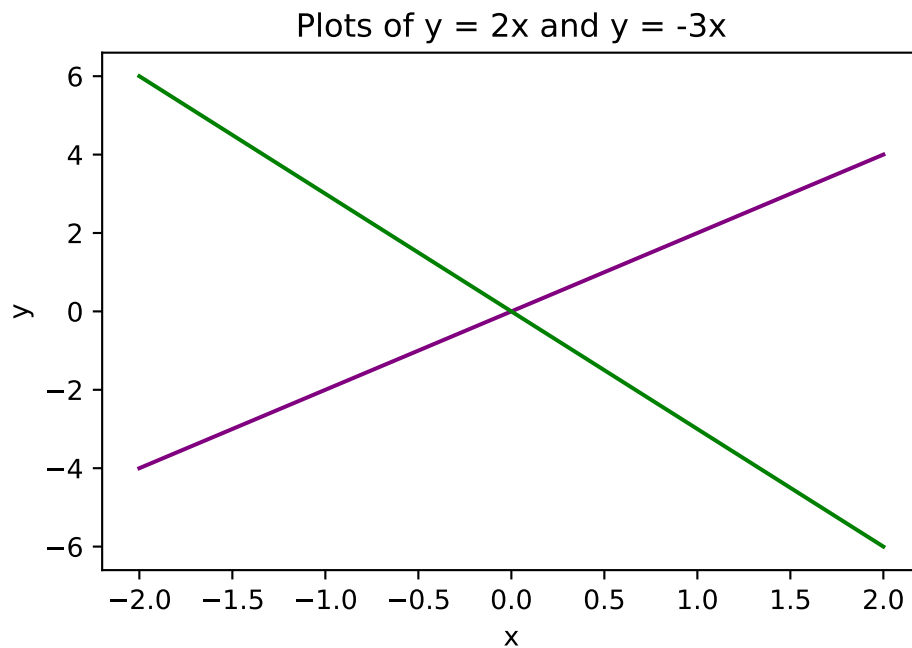
# Create line plot
plt.plot(x,y1, color = "purple")
plt.plot(x,y2, color = "green")

# Add title
plt.title("Plots of  $y = 2x$  and  $y = -3x$ ")

# Add axes labels
plt.xlabel("x")
plt.ylabel("y")

```

Text(0, 0.5, 'y')



Scatter plot

We use `plt.scatter()` to put together a scatter plot:

```

# Create sample data
x = [0, 1, 2, 3, 4, 5]
y = [0, 1, 4, 9, 16, 25]

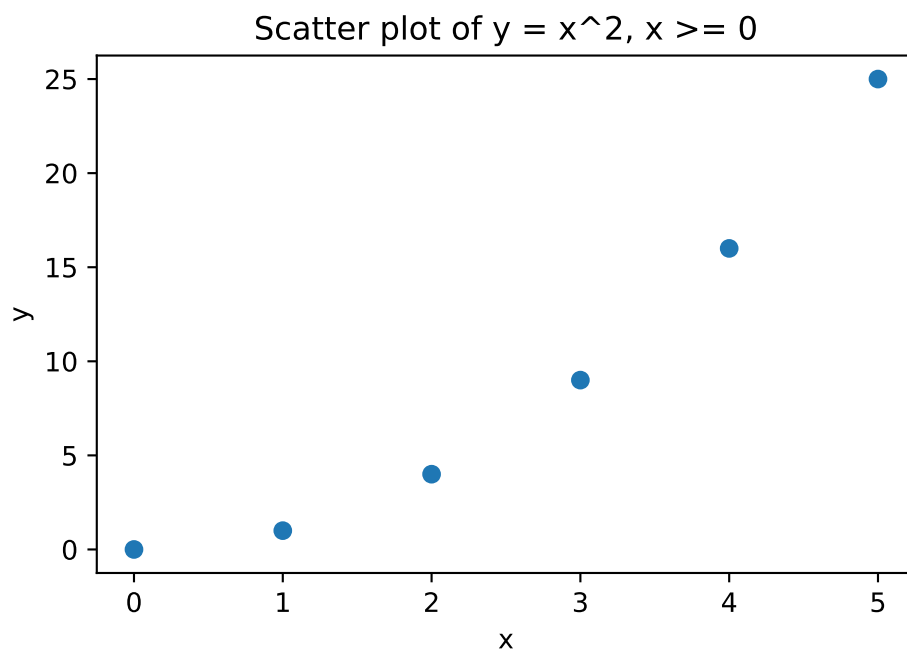
```

```
# Create scatter plot
plt.scatter(x, y)

# Add title
plt.title("Scatter plot of  $y = x^2$ ,  $x \geq 0$ ")

# Add axes labels
plt.xlabel("x")
plt.ylabel("y")
```

```
Text(0, 0.5, 'y')
```



Bar plot

We use `plt.bar()` to put together a bar plot:

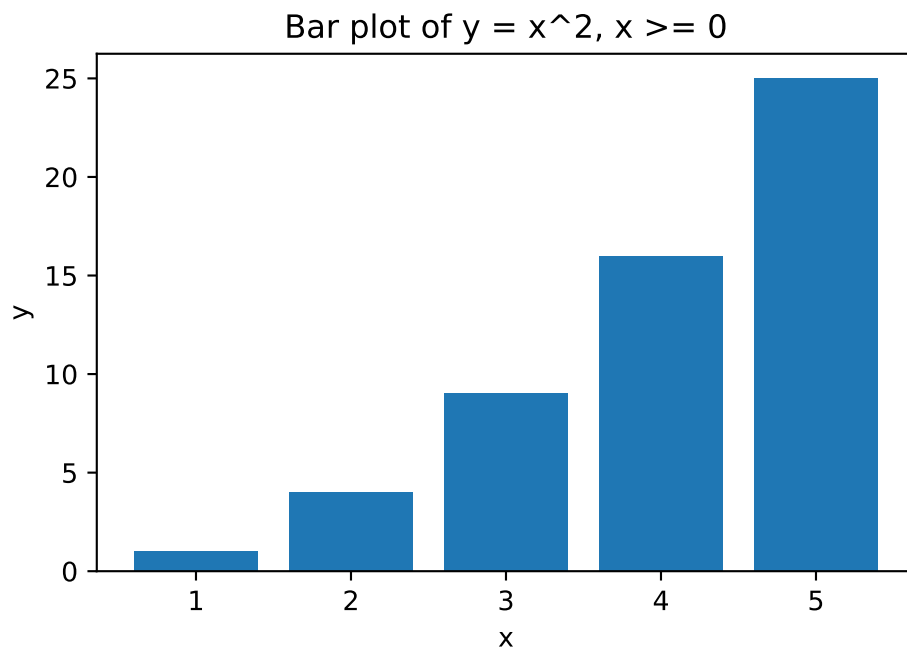
```
# Create sample data
x = [1, 2, 3, 4, 5]
y = [1, 4, 9, 16, 25]
```

```
# Create scatter plot
plt.bar(x, y)

# Add title
plt.title("Bar plot of  $y = x^2$ ,  $x \geq 0$ ")

# Add axes labels
plt.xlabel("x")
plt.ylabel("y")
```

```
Text(0, 0.5, 'y')
```



Histogram

We use `plt.hist()` to put together a histogram.

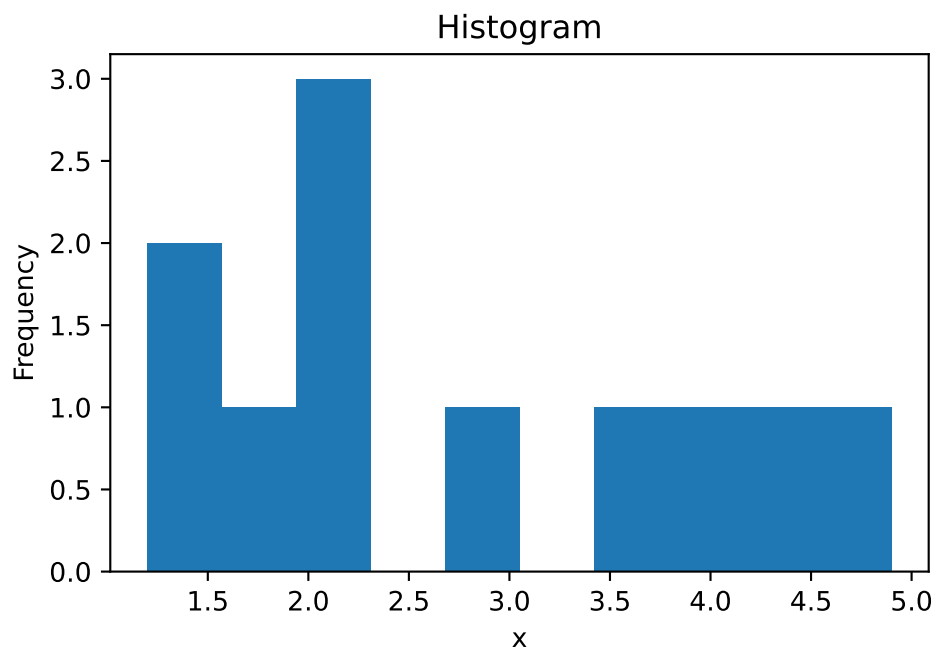
```
# Create sample data
x = [1.2, 1.5, 1.7, 2, 2.1, 2.2, 2.8, 3.6, 4.1, 4.4, 4.9]

# Create histogram
plt.hist(x)
```

```
# Add title
plt.title("Histogram")

# Add axes labels
plt.xlabel("x")
plt.ylabel("Frequency")
```

```
Text(0, 0.5, 'Frequency')
```



`plt.hist()` will automatically set the bin widths for you.

Advanced plot customisation

Histogram bin settings

While we are on the topic of histograms, let's customise the histogram we have just created, specifically in terms of the bins.

You can set the number of bins that the histogram can have using the `bins` argument in `plt.hist()`:

```

# Create sample data
x = [1.2,1.5,1.7,2,2.1,2.2,2.8,3.6,4.1,4.4,4.9]

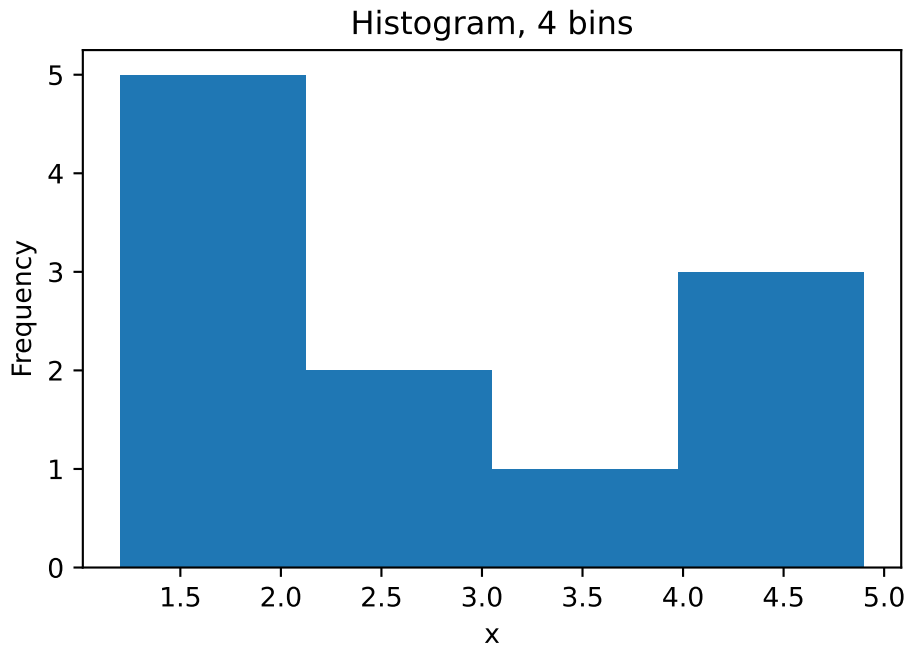
# Create histogram with 4 bins
plt.hist(x, bins = 4)

# Add title
plt.title("Histogram, 4 bins")

# Add axes labels
plt.xlabel("x")
plt.ylabel("Frequency")

```

```
Text(0, 0.5, 'Frequency')
```



Alternatively, you can set custom bin edges:

```

# Create sample data
x = [1.2,1.5,1.7,2,2.1,2.2,2.8,3.6,4.1,4.4,4.9]

# Set custom bin edges
bin_edges = [0,1.5,3,4,5]

```

```

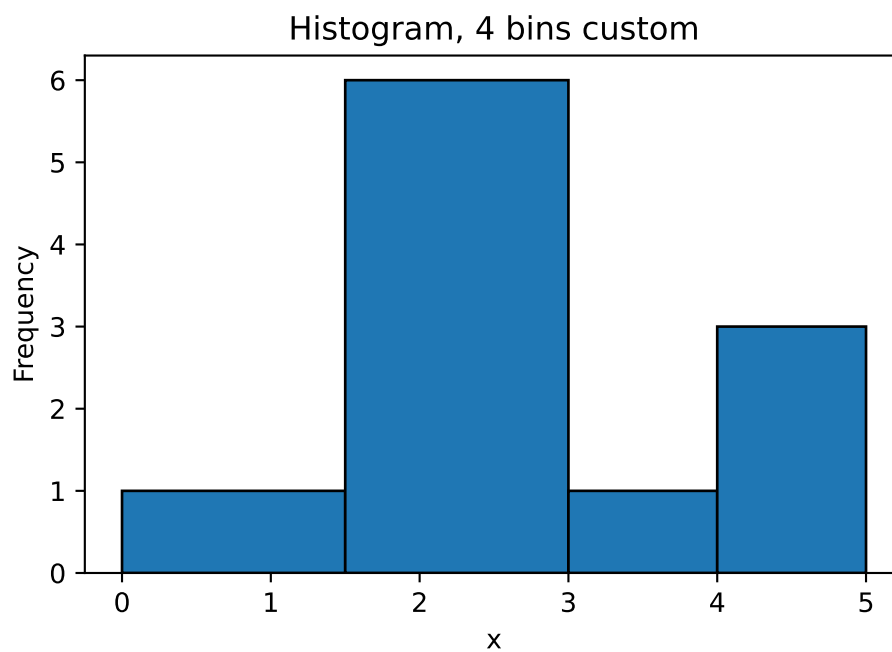
# Create histogram with 4 bins of custom width
plt.hist(x, bins = bin_edges, edgecolor = "black")

# Add title
plt.title("Histogram, 4 bins custom")

# Add axes labels
plt.xlabel("x")
plt.ylabel("Frequency")

```

```
Text(0, 0.5, 'Frequency')
```



Editing axes

Let's go back to our line plot of $y = 2x$:

```

# Create sample data
x = [-2,-1,0,1,2]
y = [-4,-2,0,2,4]

```

```

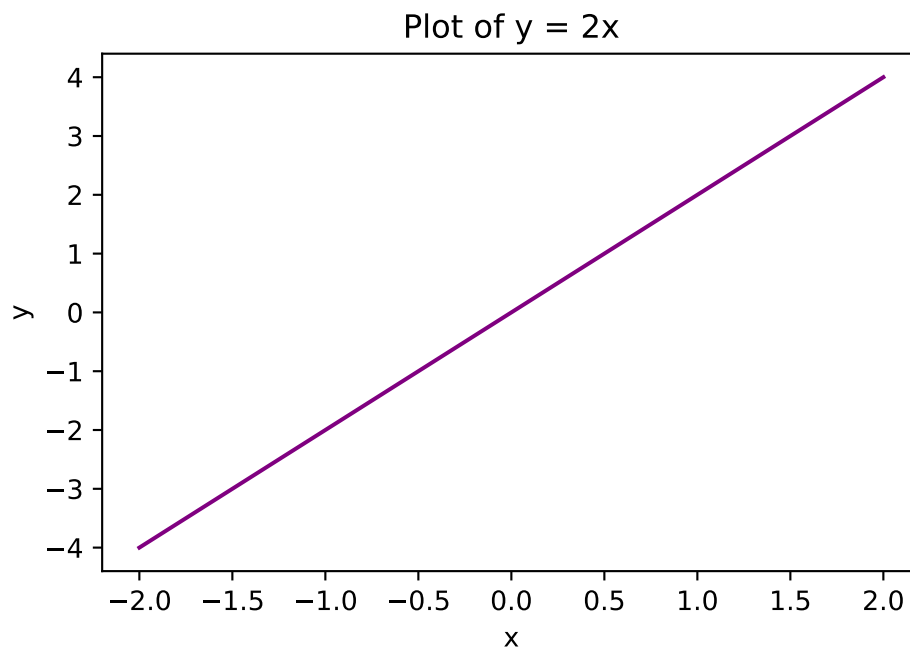
# Create line plot
plt.plot(x,y, color = "purple")

# Add title
plt.title("Plot of y = 2x")

# Add axes labels
plt.xlabel("x")
plt.ylabel("y")

```

```
Text(0, 0.5, 'y')
```



Notice that the tick marks for both the x- and y-axes are quite close together. You might prefer this as it gives you more granularity, however, some may find this quite cluttered. We can edit the axes tick marks (as well as the axes limits) using the `plt.xticks()` and `plt.yticks()` functions.

```

# Create sample data
x = [-2,-1,0,1,2]
y = [-4,-2,0,2,4]

# Create line plot

```

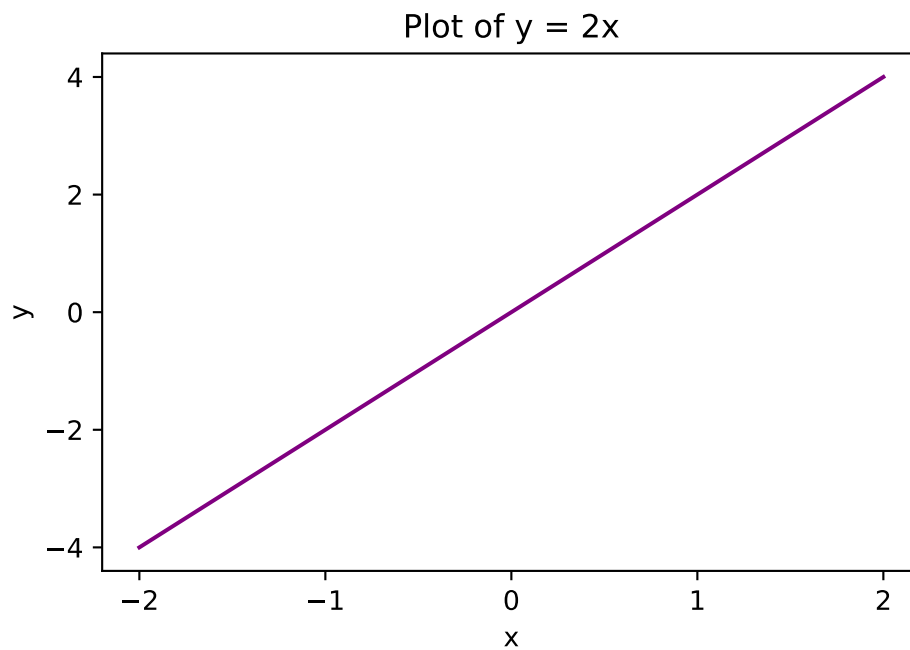
```
plt.plot(x,y, color = "purple")

# Add title
plt.title("Plot of  $y = 2x$ ")

# Edit tick marks
plt.xticks(range(-2,3))
plt.yticks([-4,-2,0,2,4])

# Add axes labels
plt.xlabel("x")
plt.ylabel("y")
```

```
Text(0, 0.5, 'y')
```



As you can see, the x- and y-axes do look significantly cleaner. We can improve how easy it is to see certain values by adding a grid using `plt.grid(True)`

```
# Create sample data
x = [-2,-1,0,1,2]
y = [-4,-2,0,2,4]
```



```

# Create line plot
plt.plot(x,y, color = "purple")

# Add title
plt.title("Plot of  $y = 2x$ ")

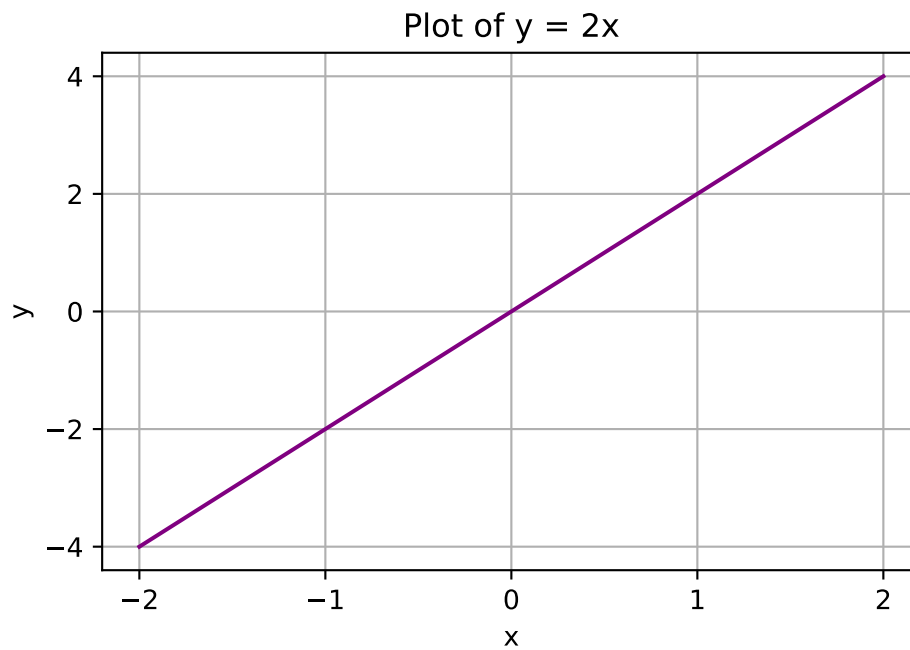
# Edit tick marks
plt.xticks(range(-2,3))
plt.yticks([-4,-2,0,2,4])

# Add grid
plt.grid(True)

# Add axes labels
plt.xlabel("x")
plt.ylabel("y")

```

```
Text(0, 0.5, 'y')
```



You can also edit the x- and y-axis limits by using `plt.xlim()` and `plt.ylim()`:

```
# Create sample data
x = [-2,-1,0,1,2]
y = [-4,-2,0,2,4]

# Create line plot
plt.plot(x,y, color = "purple")

# Add title
plt.title("Plot of y = 2x")

# Set axis limits
plt.xlim((-3,3))
plt.ylim((-5,5))

# Add axes labels
plt.xlabel("x")
plt.ylabel("y")
```

Text(0, 0.5, 'y')



Formatting text

To format the text in a plot created using Matplotlib, you can use the `fontsize` and `fontweight` arguments of the various text functions, such as `title`, `xlabel`, and `ylabel`. These arguments allow you to specify the font size and font weight (i.e. thickness) of the text, respectively.

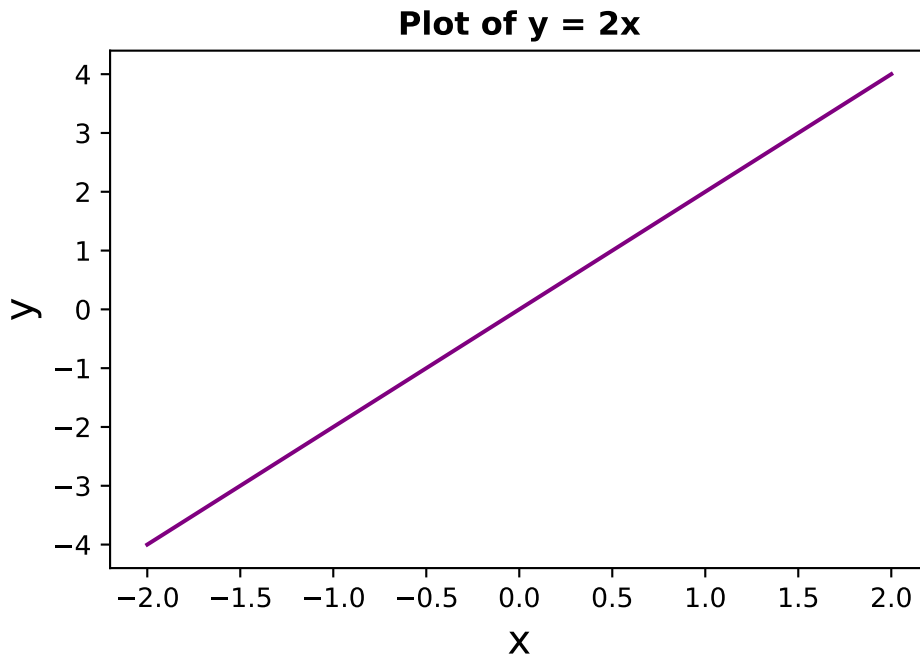
```
# Create sample data
x = [-2,-1,0,1,2]
y = [-4,-2,0,2,4]

# Create line plot
plt.plot(x,y, color = "purple")

# Add title and bold it
plt.title("Plot of  $y = 2x$ ", fontweight = 'bold')

# Add axes labels and set their font sizes to 15
plt.xlabel("x", fontsize = 15)
plt.ylabel("y", fontsize = 15)
```

`Text(0, 0.5, 'y')`



You can use the `fontstyle` argument to specify whether you would like to italicise your text. The `fontfamily` argument allows you to specify the font family, such as “serif”, “sans-serif”, or “monospace”. If you want to use a specific font, you can use the `fontname` argument instead.

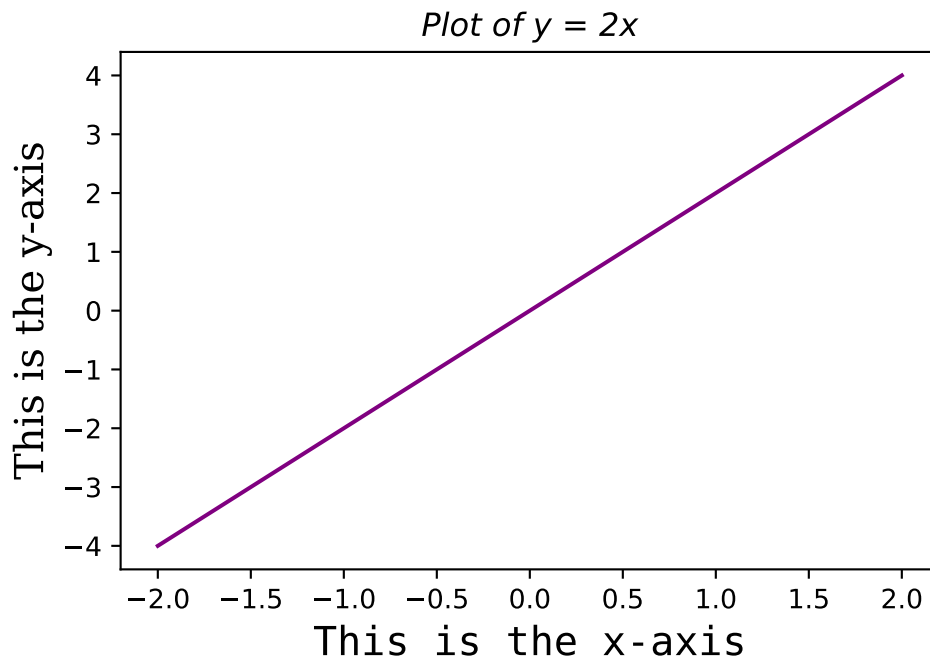
```
# Create sample data
x = [-2,-1,0,1,2]
y = [-4,-2,0,2,4]

# Create line plot
plt.plot(x,y, color = "purple")

# Add title and bold it
plt.title("Plot of  $y = 2x$ ", fontstyle = 'italic')

# Add axes labels and set their font sizes to 15
plt.xlabel("This is the x-axis", fontsize = 15, fontfamily = 'monospace')
plt.ylabel("This is the y-axis",
           fontsize = 15,
           fontfamily = 'serif')
```

```
Text(0, 0.5, 'This is the y-axis')
```



Adding a legend

You can add a legend to your plot using the `plt.legend()` argument. Notice that to label the lines in your plot, you need to use the `label` argument in the `plt.plot()` function, rather than through the `legend` function itself:

```
# Create sample data
x = [-2,-1,0,1,2]
y1 = [-4,-2,0,2,4]
y2 = [6,3,0,-3,-6]

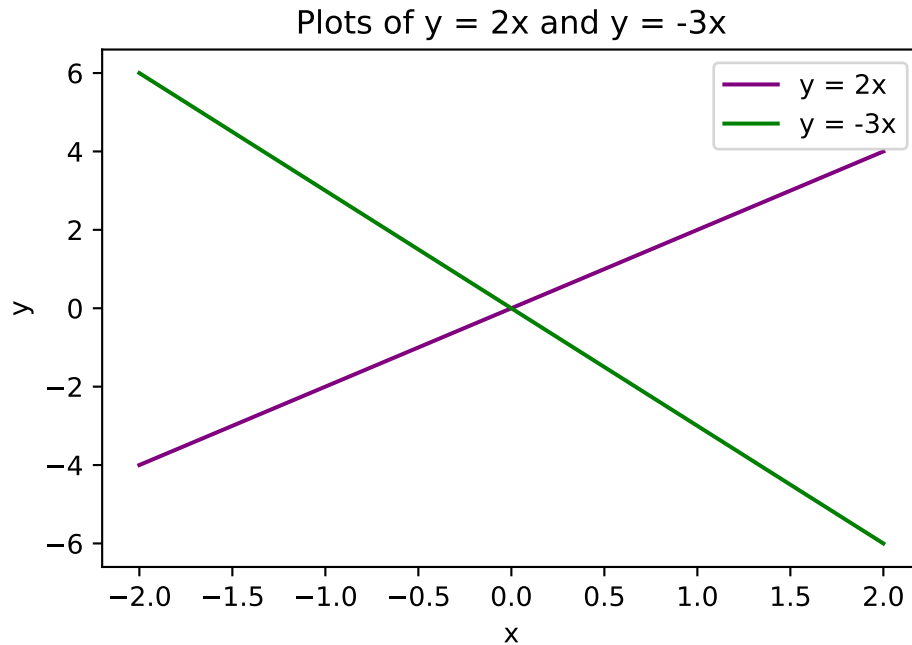
# Create line plot
plt.plot(x,y1, color = "purple", label = "y = 2x")
plt.plot(x,y2, color = "green", label = "y = -3x")

# Add title
plt.title("Plots of y = 2x and y = -3x")

# Add a legend to the top right hand corner
plt.legend(loc="upper right")

# Add axes labels
plt.xlabel("x")
plt.ylabel("y")
```

```
Text(0, 0.5, 'y')
```



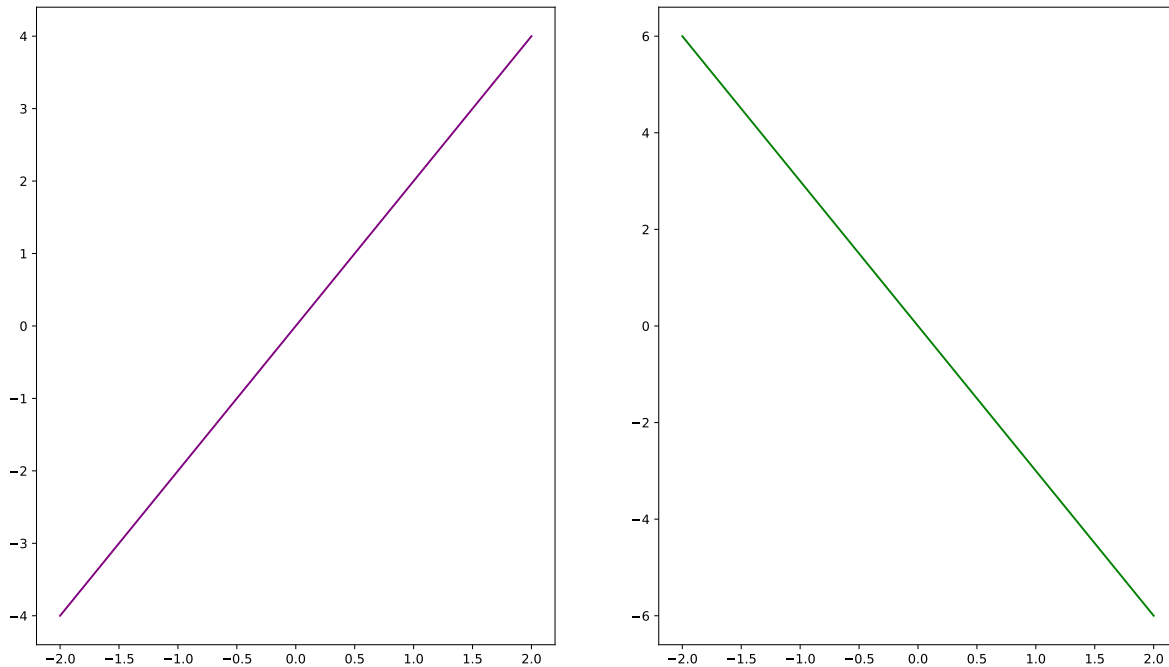
Subplots

If you want to visualise multiple plots at a time in the form of a grid, you can use the `plt.subplots()` function:

```
# Create sample data
x = [-2,-1,0,1,2]
y1 = [-4,-2,0,2,4]
y2 = [6,3,0,-3,-6]

# Create 1x2 grid of charts, with a figure size of 16x9 units
fig, ax = plt.subplots(nrows=1, ncols=2, figsize=(16, 9))

# Create a line plot in each space on the grid
ax[0].plot(x,y1, color = "purple")
ax[1].plot(x,y2, color = "green")
```



Saving plots

Use the `plt.savefig()` function to save your plots. This function takes in the name of the file that you want to save your chart to. Because of this, you can save a chart to various formats including PNG, JPEG, and TIFF.

Let's fully build our line chart and save it to `linechart.png`.

```
# Create sample data
x = [-2,-1,0,1,2]
y1 = [-4,-2,0,2,4]
y2 = [6,3,0,-3,-6]

# Create line plot
plt.plot(x,y1, color = "purple", label = "y = 2x")
plt.plot(x,y2, color = "green", label = "y = -3x")

# Add title
plt.title("Plots of y = 2x and y = -3x")

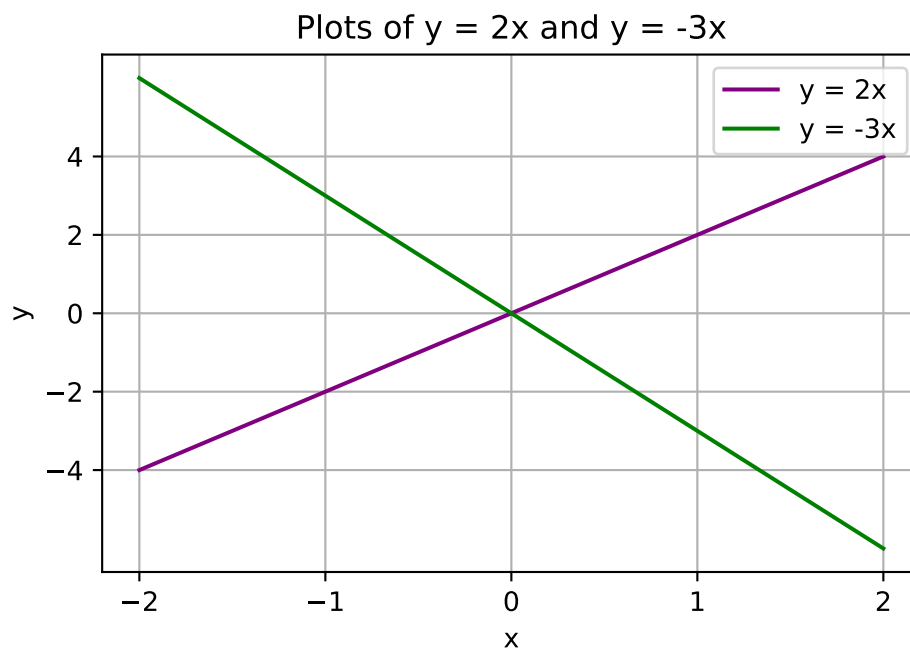
# Add a legend to the top right hand corner
plt.legend(loc="upper right")
```

```
# Edit tick marks
plt.xticks(range(-2,3))
plt.yticks([-4,-2,0,2,4])

# Add grid
plt.grid(True)

# Add axes labels
plt.xlabel("x")
plt.ylabel("y")

# Save chart
plt.savefig("linechart.png")
```



The chart should now appear in the file explorer pane in Google Colab.

That lab looks at how to incorporate Markdown and LaTeX into your Google Colab notebooks. The purpose of this part of the lab is to enhance how you annotate your code. With LaTeX, you can include equations into your notebooks.

Markdown

Markdown is a lightweight markup language that you can use to add formatting elements to plaintext text documents. It is widely used in the academic and technical communities for its simplicity and versatility. Markdown is easy to read and write, and it can be converted to HTML, PDF, and other formats.

Unfortunately there isn't one standard Markdown syntax, but the most common one is [GitHub Flavored Markdown](#). This is the syntax that Google Colab uses.

Headers

You can create headers by using the # symbol. The number of # symbols you use will determine the size of the header. For example, # Header 1 will create a large header, while ## Header 2 will create a smaller header.

Lists

You can create bulleted lists by using the - symbol. For example:

```
- Item 1  
- Item 2  
- Item 3
```

- Item 1
- Item 2
- Item 3

You can also create numbered lists by using numbers followed by periods. For example:

```
1. Item 1  
2. Item 2  
3. Item 3
```

1. Item 1
2. Item 2
3. Item 3

Emphasis

You can create emphasis by using the `*` or the `_` symbol.

- For example, `*italic*` will create *italic* text, while `**bold**` will create **bold** text.
- Equivalently, `_italic_` will create *italic* text, while `__bold__` will create **bold** text.

Links

You can create links by using the `[text](url)` syntax. For example, `[Google](https://www.google.com)` will create a link to [Google](https://www.google.com).

Images

You can include images by using the `![alt text](url)` syntax. For example, `![Google Logo](https://www.google.com/images/branding/googlelogo/1x/googlelogo_color_272x92dp.png)` will display the Google logo:



Figure 2: Google Logo

Code

You can include code snippets by using the backtick symbol (```). For example, ``print("Hello, World!")`` will display `print("Hello, World!")`.

You can also create code blocks by using three backticks. For example:

```
“python print(“Hello, World!”) “
```

will display:

```
print("Hello, World!")
```

Tables

You can create tables by using the `|` symbol to separate columns and `-` symbols to separate the header row from the content rows. For example:

```
| Header 1 | Header 2 | Header 3 |  
|-----|-----|-----|  
| Row 1, Col 1 | Row 1, Col 2 | Row 1, Col 3 |  
| Row 2, Col 1 | Row 2, Col 2 | Row 2, Col 3 |
```

will display:

Header 1	Header 2	Header 3
Row 1, Col 1	Row 1, Col 2	Row 1, Col 3
Row 2, Col 1	Row 2, Col 2	Row 2, Col 3

LaTeX

You can include LaTeX equations in your Markdown documents.

LaTeX is a typesetting system commonly used in research and other technical fields. It enables users to create high-quality documents with professional-looking mathematical and scientific equations, figures, and tables.

LaTeX has the ability to do all of the things that Markdown can do, and more. However, LaTeX has a steeper learning curve than Markdown, and it is not as widely used outside of academia and technical fields. Nowadays, its unique selling point is its mathematical typesetting capabilities — which is why Markdown's equations use LaTeX syntax.

For full LaTeX documents, you can install a desktop-based distribution such as [MikTeX](#), or online using [Overleaf](#). However, in this lab we will just be looking at how you can incorporate LaTeX into Google Colab.

Inline equations and display mode

There are two ways you can incorporate LaTeX equations: either by using the `$$` notation or by using the `\[ \]` notation.

By wrapping your equation in one dollar sign (`$`), you can write mathematical expressions in-line. For example, `$E = mc^2$` becomes $E = mc^2$.

By wrapping your equation in two dollar signs (\$\$), you can write mathematical expressions in “display mode”, which puts expressions on a standalone line: `$$a^2 + b^2 = c^2.$$` becomes

$$a^2 + b^2 = c^2.$$

You can also create mathematical expressions in display mode by wrapping your expression in `\[` and `\]` symbols. However, this does not seem to work in Google Colab for now, so we recommend using the dollar sign notation instead.

Mathematical notation

Use braces `{}` if there are multiple terms in the exponent: `$x^{a+b}$` becomes x^{a+b} .

Inside a math environment, superscripts are denoted with `(^)` and subscripts denoted with `(_)`. For example, `$a^b$` and `$a_b$` become a^b and a_b respectively.

There are many mathematical symbols that can be called upon using the backslash followed by the name of the symbol, including Greek symbols.

`\alpha`, `\beta`, `\gamma`, `\delta`, `\pi`, `\Pi`, `\phi`, `\Phi`

$\alpha, \beta, \gamma, \delta, \pi, \Pi, \phi, \Phi$

There are far too many to list, and you can find a comprehensive list at [The Comprehensive LaTeX Symbol List - The CTAN archive](#)

LaTeX has an equivalent of ‘functions’ (commands) which also start with backslash but take one or more arguments in braces. For example, take the following equation for the area of a circle:

$$\text{Area of Circle} = \pi r^2$$

This equation is represented by the following LaTeX code:

```
$$\text{Area of Circle} = \pi r^2$$
```

The `\text{}` command converts the text in the expression from italicised to non-italicised.

Other commands include `\frac{}{}` (which requires two arguments), `\sqrt{}`, and `\partial`:

```
$$\frac{\partial}{\partial x} = \sqrt{x}$$
```

$$\frac{\partial}{\partial x} = \sqrt{x}$$

Matrices

For matrices with square brackets (braces), use `\begin{bmatrix}` `\end{bmatrix}`

```
$$B = \begin{bmatrix}
a & b & c \\
d & e & f \\
g & h & i
\end{bmatrix}$$
```

$$B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

For matrices with parentheses, use `\begin{pmatrix}` `\end{pmatrix}`

```
$$\sigma^2 = \begin{pmatrix}
\sigma_1^2 & \sigma_{12} \\
\sigma_{12} & \sigma_2^2
\end{pmatrix}$$
```

$$\sigma^2 = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}$$

General matrix notation (notice the `cdots`, `vdots`, and `ddots`) :

```
$A_{m,n} =
\begin{pmatrix}
a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\
a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\
\vdots & \vdots & \ddots & \vdots \\
a_{m,1} & a_{m,2} & \cdots & a_{m,n}
\end{pmatrix}
```

$$A_{m,n} = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix}$$

Examples from last week's Lab

- $\mathbf{z} = \mathbf{X}\mathbf{w} + \mathbf{b}$ produces $\mathbf{z} = \mathbf{X}\mathbf{w} + \mathbf{b}$,
- Softmax function: $\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$ produces $\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$.

RMarkdown and Quarto

The Jupyter Notebook experience separates code cells from Markdown cells. It is more convenient for experimenting with code, and tasks where there is more coding than there is writing. However, for tasks where there is more writing than coding, it is more convenient to use RMarkdown or Quarto.

RMarkdown is a variant of Markdown that allows you to include R code chunks in your document. It runs the R code and includes the output in the document. It has been around for many years, and I recommend watching Rob Hyndman's presentation on how he uses RMarkdown for nearly all his writing tasks.

https://youtu.be/_D-ux3MqGug?si=ZTxJuAfUPpuS0yKA

Quarto is a newer improvement on RMarkdown that supports Python, R, and many other programming languages, and improves upon RMarkdown in a number of ways.

To create a code chunk which will be executed, you can use the following syntax:

```
```{python}
#| label: fig-polar
#| fig-cap: "A line plot on a polar axis"

import numpy as np
import matplotlib.pyplot as plt

r = np.arange(0, 2, 0.01)
theta = 2 * np.pi * r
fig, ax = plt.subplots(
 subplot_kw = {'projection': 'polar'})
```

```

)
ax.plot(theta, r)
ax.set_rticks([0.5, 1, 1.5, 2])
ax.grid(True)
plt.show()
...

```

This produces:

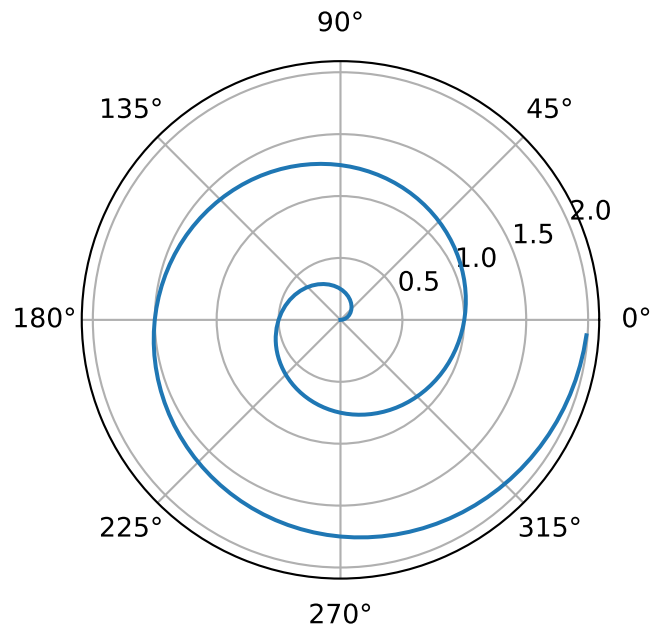


Figure 3: A line plot on a polar axis

Quarto is the software I used to make this website, the lecture slides, my personal website (<https://laub.au/>) and much more. While I don't particularly recommend you use Quarto for your assignments, I do recommend you use it for your personal projects.

## Acknowledgements

Thanks to Sam Luo & Eric Dong who contributed to the LaTeX section of the lab.

I used GitHub Copilot to draft the earlier Markdown demonstrations.